

EXPERIMENT- 2

(A)AIM:

To design a circuit and write a program to interface a 16x2 LCD in 4-bit mode with Arduino Uno, and display the message "UIC [North Campus]" on the LCD.

OBJECTIVE:

To design and implement a circuit for interfacing a 16×2 LCD with Arduino Uno in 4-bit mode and write a program to display the message “UIC [North Campus]” on the LCD.

APPARATUS / COMPONENTS REQUIRED:

S.No.	Component	Quantity
1	Arduino Uno board	1
2	16x2 LCD display (HD44780 compatible)	1
3	10K ohm potentiometer (contrast control)	1
4	220 ohm resistor (backlight current limiting)	1
5	Breadboard	1
6	Connecting / jumper wires	as required
7	USB cable (for programming Arduino)	1

THEORY:

A Liquid Crystal Display (LCD) is commonly used to display status or data in embedded systems. A 16x2 LCD means it can display 16 characters per line, over 2 lines. Most 16x2 LCDs are based on the HD44780 driver chip, which can operate in two modes: 8-bit mode (uses 8 data lines D0-D7) and 4-bit mode (uses only 4 data lines D4-D7).

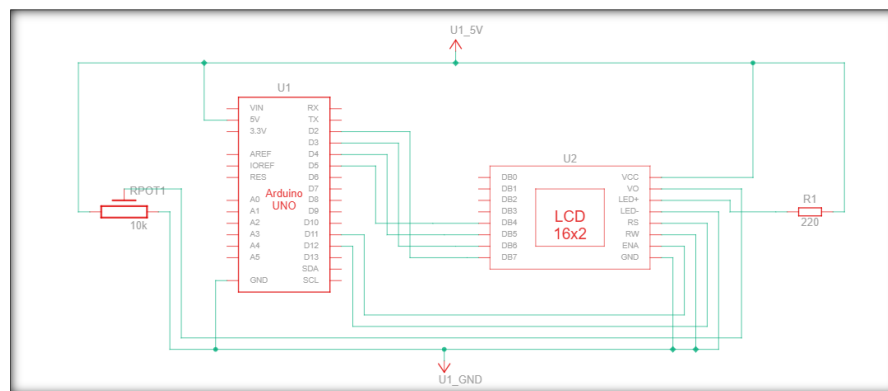
4-bit and 8-bit Mode of LCD:

The LCD can work in two different modes, namely the 4-bit mode and the 8-bit mode. In 4-bit mode, we send the data nibble by nibble, first the upper nibble and then the lower nibble. For those of you who don't know what a nibble is: a nibble is a group of four bits, so the lower four bits (D0-D3) of a byte form the lower nibble while the upper four bits (D4-D7) of a byte form the higher nibble. This enables us to send 8-bit data.

Whereas in 8-bit mode, we can send the 8-bit data directly in one stroke since we use all 8 data lines.

Now you must have guessed it, yes, 8-bit mode is faster and flawless than 4-bit mode. But the major drawback is that it needs 8 data lines connected to the microcontroller. This will make us run out of I/O pins on our MCU, so 4-bit mode is widely used. No control pins are used to set these modes. It's just the way of programming that changes.

4-bit mode is preferred in microcontroller projects because it uses fewer I/O pins of the Arduino, freeing them for other tasks, while sending each byte of data as two nibbles (upper nibble first, then lower nibble).



PIN CONNECTION TABLE:

LCD Pin	Function	Connected To
1	VSS	Arduino GND
2	VDD	Arduino 5V
3	V0 (Contrast)	Wiper of 10K potentiometer
4	RS	Arduino Digital Pin 12
5	RW	Arduino GND
6	EN	Arduino Digital Pin 11
7 to 10	D0 - D3	Not Connected (4-bit mode)
11	D4	Arduino Digital Pin 5
12	D5	Arduino Digital Pin 4
13	D6	Arduino Digital Pin 3
14	D7	Arduino Digital Pin 2
15	A (Backlight +)	Arduino 5V (through 220 ohm resistor)
16	K (Backlight -)	Arduino GND

PROCEDURE:

1. Connect the 16x2 LCD to the Arduino Uno as per the circuit diagram and pin connection table shown above.
2. Connect the wiper terminal of the 10K potentiometer to the V0 pin of the LCD, and its outer two terminals to 5V and GND, to adjust display contrast.
3. Connect a 220-ohm resistor in series with the backlight anode (pin 15/A) to 5V, and the backlight cathode (pin 16/K) to GND.
4. Open the Arduino IDE and write / paste the program given above.

5. Select the correct board (Arduino Uno) and COM port from the Tools menu.
6. Compile the program (Sketch > Verify/Compile) and check for errors.
7. Upload the program to the Arduino Uno board (Sketch > Upload).
8. Adjust the potentiometer knob until the characters are clearly visible on the LCD.
9. Observe the message "UIC" on the first line and "[North Campus]" on the second line of the LCD.

CODE:

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
```

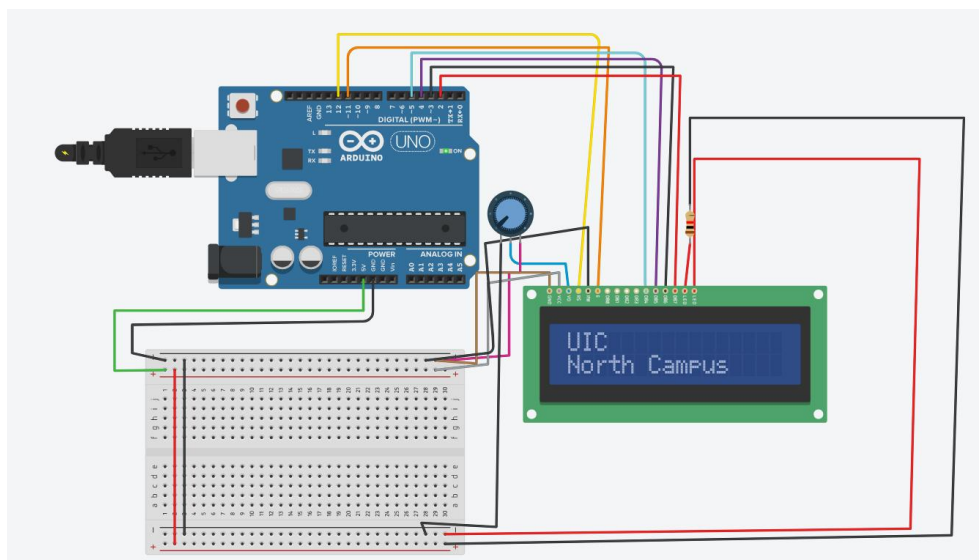
```
void setup() {
  lcd.begin(16, 2);

  lcd.setCursor(0, 0);
  lcd.print("UIC");

  lcd.setCursor(0, 1);
  lcd.print("North Campus");
}
```

```
void loop() {
}
```

CIRCUIT DIAGRAM:



OBSERVATION TABLE:

(a) Control signal states observed during operation:

S.No.	RS	R/W	EN	Operation Performed
1	0	0	Pulse (High to Low)	Send command (e.g. lcd.begin, lcd.clear)
2	1	0	Pulse (High to Low)	Send data / character to be displayed

(b) Message display observation:

S.No.	LCD Line	Cursor Position (Col, Row)	Message Sent	Output Observed on LCD	Result
1	Line 1	(0, 0)	UIC	UIC	As expected
2	Line 2	(0, 1)	[North Campus]	[North Campus]	As expected

RESULT:

The 16x2 LCD was successfully interfaced with the Arduino Uno in 4-bit mode, and the message "UIC [North Campus]" was displayed correctly on the LCD screen.

PRECAUTIONS:

1. Ensure correct pin connections between LCD and Arduino before powering the circuit.
2. Do not exceed the voltage rating of the LCD (operates at 5V).
3. Always connect a current-limiting resistor (220 ohm) in series with the backlight pin.
4. Adjust the contrast potentiometer slowly to avoid a blank / invisible display.
5. Double-check the RW pin is grounded, since the library assumes write-only operation.

CONCLUSION:

The 16x2 LCD was successfully interfaced with the Arduino Uno in 4-bit mode, using only six control/data lines (RS, E, D4–D7) instead of the ten required in 8-bit mode. This confirmed that 4-bit mode is an efficient choice for microcontroller-based systems with limited I/O pins, at the minor cost of slightly slower (nibble-wise) data transfer.

Using the built-in LiquidCrystal.h library, the Arduino was programmed to initialize the display and correctly output the custom message "UIC [North Campus]" on the LCD screen.

(B)AIM:

To design a circuit and write a program to interface a 16x2 LCD using I2C communication protocol with Arduino Uno, and display the message "CHANDIGARH UNIVERSITY" on the LCD.

OBJECTIVE:

To design and implement a circuit for interfacing a 16×2 LCD with Arduino Uno using the I2C communication protocol and write a program to display the message “CHANDIGARH UNIVERSITY” on the LCD.

APPARATUS REQUIRED:

S. No.	Component	Quantity
1	Arduino Uno Board	1
2	16x2 LCD with I2C Adapter Module (PCF8574)	1
3	Connecting Wires (Male-Female)	4
4	USB Cable (for Arduino programming)	1
5	Breadboard (optional)	1
6	Personal Computer with Arduino IDE installed	1

THEORY:

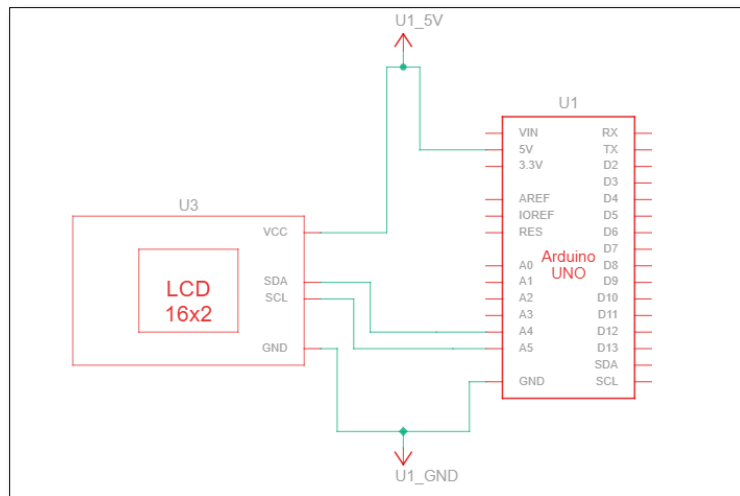
The 16x2 LCD requires multiple I/O pins (RS, EN, D4-D7) when connected directly in 4-bit mode, which occupies several digital pins on the Arduino. To overcome this, an I2C (Inter-Integrated Circuit) adapter module based on the PCF8574 I/O expander chip is mounted on the back of the LCD. This module converts the parallel LCD interface into a serial I2C interface, requiring only two wires (SDA and SCL) for communication, along with power supply lines (VCC and GND).

I2C is a synchronous, two-wire serial communication protocol consisting of:

- SDA (Serial Data Line) - carries the data
- SCL (Serial Clock Line) - carries the clock signal

CONNECTION TABLE:

PCF8574 Pin (I2C Module Pin)	Arduino Uno Pin	Function
GND	GND	Ground connection
VCC	5V	Power supply
SDA	A4	Serial Data Line
SCL	A5	Serial Clock Line



LIBRARY REQUIRED:

Before running the program, install the following libraries via Arduino IDE -> Sketch -> Include Library -> Manage Libraries:

- LiquidCrystal_I2C
- Wire (built-in, no installation needed)

CODE:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x20, 16, 2);
void setup()
{
  Wire.begin();
  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("CHANDIGARH");
  lcd.setCursor(0, 1);
  lcd.print("UNIVERSITY");
}
void loop()
{
}
```

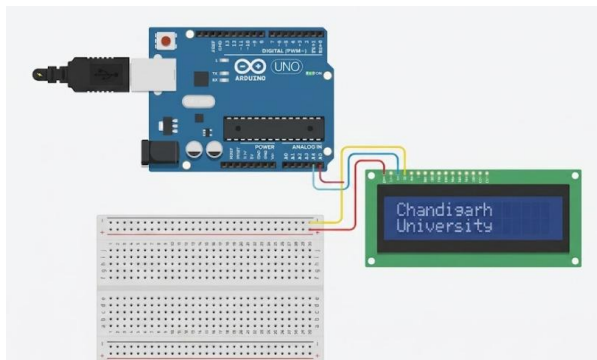
Explanation of Functions Used:

Function/Command	Purpose
<code>#include <LiquidCrystal_I2C.h></code>	Includes the library required to control an I2C LCD.
<code>LiquidCrystal_I2C lcd(0x20,16,2);</code>	Creates an LCD object with I2C address 0x20, 16 columns, and 2 rows.
<code>setup()</code>	Executes only once when the Arduino starts or resets.
<code>lcd.init();</code>	Initializes the I2C LCD for communication.
<code>lcd.backlight();</code>	Turns on the LCD backlight for better visibility.
<code>lcd.setCursor(column, row);</code>	Moves the cursor to the specified column and row.
<code>lcd.print("Text");</code>	Displays the specified text on the LCD at the current cursor position.

PROCEDURE:

1. Mount the I2C adapter module onto the back of the 16x2 LCD (solder if not pre-attached).
2. Connect VCC, GND, SDA, and SCL pins of the I2C module to the Arduino Uno as per the connection table.
3. Connect the Arduino Uno to the computer using the USB cable.
4. Open Arduino IDE and install the LiquidCrystal_I2C library.
5. Upload the I2C Scanner sketch first to determine the correct I2C address of the module.
6. Replace the address in the main program (if different from 0x27) and upload the final program.
7. Observe the LCD screen for the displayed message.
8. Adjust the contrast potentiometer on the I2C module (if present) for clear visibility of characters.

OUTPUT:



OBSERVATION:

On successful upload of the program, the LCD screen displayed:



The backlight turned on, and the text appeared clearly and legibly on both rows of the display.

RESULT:

The 16x2 LCD was successfully interfaced with the Arduino Uno using the I2C communication protocol, and the message "CHANDIGARH UNIVERSITY" was displayed correctly on the screen using only 4 connecting wires.

CONCLUSION:

In this experiment, a 16x2 LCD display was interfaced with an Arduino Uno using the I2C protocol via a PCF8574-based I2C adapter module. Unlike the conventional 4-bit or 8-bit parallel interfacing methods that require 6 to 10 connections, the I2C method simplified the wiring to just four wires (VCC, GND, SDA, SCL), significantly reducing circuit complexity and freeing up Arduino's digital pins for other peripherals.

PRECAUTIONS:

1. Ensure the LCD's I2C module is firmly connected and soldered properly.
2. Do not interchange SDA and SCL connections.
3. Verify the I2C address using the scanner sketch before running the main program.
4. Adjust the potentiometer on the I2C module gently to avoid damaging it while setting contrast.
5. Ensure Arduino is properly powered via USB before uploading the program.